

In T.617/07 the claim at issue concerned monoclonal antibodies and synthetic and biotechnological derivatives thereof defined by structural and functional features. The board found that, given his common general knowledge, the skilled person would be able, in a possibly time-consuming but straightforward manner, to provide antibody variants having the functional requirements indicated in the claim. There was no doubt that the structural definition in the claim included antibodies that did not have the desired function but, when attempting to rework the invention the skilled person would on the basis of his knowledge be able to avoid non-functional variants. Therefore, because the skilled person knew how to achieve antibodies with the desired function on the basis of a particular known antibody, he was not in the situation of having to sort out non-functional variants in a burdensome manner.

In T.386/08 the patent concerned humanised antibodies with framework sequences. It disclosed not only one, but many examples. The board pointed out that the concept of sufficiency of disclosure over the whole scope of the claim did not mean that, for a disclosure to be considered as sufficient, it had to be demonstrated that each and every conceivable embodiment of a claim could be obtained; see G.1/03 (OJ 2004, 413). There may be situations where the specification contains sufficient information on the relevant criteria for finding appropriate alternatives ("variants") over the claimed range with reasonable effort. Under these circumstances the non-availability of certain variants encompassed by the claim at the priority date is considered immaterial for the sufficiency of disclosure. For an example where this was not so, see T.601/05. The current situation however was different in that the patent described quite a number of appropriate alternatives and in that the allegedly non-obtainable variants were "hypothetical" variants. The requirements of Art. 83 EPC were fulfilled.

Art. 83 EPC was not complied with in T.941/16 (anti-PSMA antibody). The board decided that, in the absence of any examples of a claimed antibody/fragment, the general information in the patent application and the common general knowledge, taken together, could not be considered to provide the information necessary to allow the skilled person to reliably obtain substantially all of the claimed antibodies/fragments fulfilling the functional requirements of the claim. For particular combinations of complementarity-determining regions (CDRs) it was not credible that a humanised antibody/fragment with the properties defined in claim 1 would be obtained. Readily performing the invention across the entire scope of the claim placed an undue burden on the skilled person. Lastly, the board, with reference to G.1/03 (OJ 2004, 413, point 2.5.2 of the Reasons), dealt with the argument made by the appellant (applicant) that patent applications in the field of biochemistry should not be treated worse than those in the other fields of classical chemistry (presence of **non-working embodiments** in a generic chemical formula).

The March 2022 version of the Guidelines includes a new section on antibodies since 2021 (see G-II, 5.6). As regards claims directed to antibodies defined by **functional features** (i.e. by their ability to perform a certain function), they state in G-II, 5.6.1.3: "If an antibody is defined exclusively by functional properties, it has to be carefully assessed whether the application provides an enabling disclosure across the whole scope claimed". As to the related case law, the boards held Art. 83 EPC not to be met in the following

decisions: T. 1466/05, T. 601/05, T. 1389/13. For examples of cases in which sufficient disclosure was established, see T. 2045/09 and T. 845/19 (antibody defined, after limitation, by structural and functional features).

For another decision concerning antibodies, see T. 32/17 (definition of an antibody by reference to a deposited hybridoma – R. 31 EPC).

Whether opponents can cite post-published documents

In T. 1872/16 (claim 1 was drafted in the form of a purpose-limited product claim pursuant to Art. 54(5) EPC) the suitability of an IL-13 antibody as defined in the claim for the treatment of specific forms of asthma had to be assessed. The application did not contain any experiments testing the suitability of the antibodies and did not disclose a link between IL-13 and the specific forms of asthma recited in the claim. The appellant (patent proprietor) disputed that the respondents-opponents could use post-published documents (i.e. D23, post-published clinical trial) to refute the suitability of a claimed compound for a claimed therapeutic application, submitting that only documents available at the effective date of the patent could be taken into account. The board explained that the standard established in the case law was that serious doubts had to be substantiated by verifiable facts. The opponents were free to use whatever evidence they chose to substantiate those serious doubts. This also applied to the date on which the evidence was generated. There might be cases where the serious doubts could only be properly substantiated by facts which were obtained after the effective date of the patent (see T. 219/01 in relation to a medical use claim and lack of sufficient disclosure on the basis of post-published clinical trial). The respondents could therefore base their arguments on document D23. In the board's view, with the patent not containing any examples of antibodies which had been shown to be suitable for treating severe asthma, it was sufficient for the respondents to show one embodiment which was not suitable. On the basis of D23 the skilled person would have had serious doubts as to the suitability of the antibody for treating severe asthma. The requirements of Art. 83 EPC were not met.

7.4. Factors contributing to a finding of undue burden

In T. 187/93 there were experimental uncertainties in the patent application. The board found that the skilled person, when trying to obtain the same technical effect with a different glycoprotein would have experienced lack of predictability, which amounted to an undue burden.

In T. 2006/08, although no experimental details were provided for factor IX in the patent-in-suit, the board considered that no undue experimentation would be required to carry out the method steps. It was plausible that the claimed process achieved an improvement of the in vivo function of factor IX. The requirements of Art. 83 EPC were fulfilled.

Similarly, in T. 727/95, the board found that the invention relied too much on chance. The claimed subject-matter included a "microorganism designated Acetobacter and having the ability of microorganisms [...]". The board observed that by including the phrase "having the ability of", the claim covered not only Acetobacter microorganisms derived from the

deposited strains, but also *Acetobacter* microorganisms which had the stated characteristics in common with the deposited strains. In the board's judgment, finding other stable, cellulose high-producing *Acetobacter* strains in nature was a chance event, and relying on chance for reproducibility amounted to an undue burden in the absence of evidence that such chance events occurred and could be identified frequently enough to guarantee success. The board concluded that the claim was not repeatable without undue burden over the entire breadth of the claim.

The claimed subject-matter in **T. 639/95** concerned a method for producing PHB biopolymers in a host transformed with genes encoding the enzymes β -ketothiolase, acetoacetyl-CoA reductase and polyhydroxy butyrate (PHB) synthetase. The board found that the experimental plan for identifying and isolating the PHB gene was very general. Some references were missing and/or incomplete. There were no results and no details which could facilitate the repetition of the work. The board thus held that the total amount of experimental effort necessary amounted to an undue burden for the skilled person.

However, in **T. 412/93**, where errors and omissions prejudiced the reproducibility of one of the examples in toto and of another example in part, the reproducibility of the invention was not affected, as the examples were alternatives to previous ones.

In **T. 612/92**, further scientific research would have been necessary in order to carry out the invention in some of the areas claimed. The board held the requirements of Art. 83 EPC 1973 were not fulfilled because there were serious doubts as to whether such a method could be performed over the whole range that was claimed (see **T. 694/92**, OJ 1997, 408).

However, in **T. 223/92** the disclosure enabled those skilled in the art to reproduce the invention, possibly in a time-consuming and cumbersome way, but, in the given circumstances, without undue burden of experimentation and without needing inventive skill (see also **T. 412/93**).

T. 1456/06 concerned the level of disclosure required for enablement of a claim directed to peptide vaccines. It was apparent from the prior art that the development of peptide-based vaccines to treat cancer – the sole specific type of vaccine mentioned in the application as filed – was not only extremely laborious, but also fraught with uncertainties. The application as filed did not disclose any telomerase peptide which might – plausibly – be regarded as a suitable candidate for a vaccine, nor did it contain either technical information as to how to identify possible candidate peptides, or instructions on how to proceed in case of failure. The board concluded that identifying immunogenic fragments of the telomerase protein suitable for the manufacture of a vaccine by a trial and error procedure constituted an undue burden to a person skilled in the art.

The application in **T. 1364/08** concerned viruses for the treatment of cellular proliferative disorders. It provided no experimental data proving that the claimed adenovirus was able to replicate in cells having an activated Ras-pathway but not in normal cells. No data was present demonstrating that such a virus could be useful for the treatment of Ras-mediated cell proliferative disorders. However, based on what was described in the application as

filed and taking into account what was known in the prior art, it was credible that the modified adenovirus specified in the claim would have been effective for the treatment of Ras-mediated cell proliferative disorder. Post-published evidence could therefore be taken into account to back up this evidence (following T 609/02).

In T 1846/10 the invention under consideration related to the preparation of a live vaccine against *L. intracellularis* which relied on the use of attenuated *L. intracellularis* bacteria. *L. intracellularis* is the causative agent of proliferative enteropathy in pigs, also known as porcine proliferative enteropathy (PPE). To be suitable as a live vaccine strain, the attenuated bacteria must fulfil the following three criteria: (i) apathogenicity, which means that they do not cause the disease; (ii) be suitable and retain immunogenicity, which means that they induce protective immunity in the animal host, and (iii) genetic stability, which means that they do not revert to being pathogenic or conversely become too attenuated. This was undisputed among the parties.

The skilled person, wanting to carry out the claimed invention, could not rely on his common general knowledge or the prior art to obtain suitable attenuated *L. intracellularis* bacteria. The patent had to provide the necessary guidance for the successful implementation of the claimed invention. The board concluded that the guidance provided by the patent did **not** allow the skilled person to obtain an attenuated *L. intracellularis* strain **without undue burden** or inventive step.

The skilled person was taught by the patent that he had to test the passaged strain, but only to confirm attenuation. The *L. intracellularis* strain suitable to carry out the invention not only had to be less virulent than the corresponding wild type strain, but also had to fulfil the additional two criteria of appropriate immunogenicity and genetic stability. Relying on the guidance provided by the patent and not knowing why the strain used in example 5 did not protect the vaccinated animals he would have had no reason to assume that the number of passages had to be increased. The person skilled in the art would not be inclined to consider the intervals disclosed in the description of the patent as mere lower limits but would have understood these indications as concrete ranges. The board concluded that example 5 of the patent represented evidence that the skilled person, by following the guidance of the patent, would fail to obtain an attenuated strain of *L. intracellularis* suitable for the preparation of a live vaccine.

In T 1376/11 the board concluded that the only way disclosed in the application to arrive at the paprika plants of the invention started from parental *Capsicum annuum* NM varieties, such as *Capsicum annuum* NM 1441. The public availability of these parental plants at the priority date of the application was therefore a mandatory requirement for the skilled person to reproduce the invention. In the absence of evidence that *Capsicum annuum* NM 1441 was publicly available, the board concluded that the application did not disclose the subject-matter of claim 1 in a manner sufficiently clear and complete for it to be carried out by the skilled person.

7.5. Requirements relating to nucleotide and amino acid sequences

Under R. 30(3) EPC, where the applicant has not filed the necessary sequence listing prescribed in R. 30(1) EPC, the EPO shall invite him to furnish it on payment of a fee. In J. 7/11, the Legal Board held that such invitations cannot be given orally only – a telephone call was, given the short time available, useful, but had to be followed by a written invitation enumerating all objections raised. Failure to do so amounted to a substantial procedural violation.

In J. 8/11 the crucial issue on appeal was the interpretation of the term "disclosed" in R. 30(1) EPC, namely the question whether a patent application which related to the use of polypeptides well known in the prior art and which identified these polypeptides by their common names and by database accession numbers concerning specific representative sequences had to be regarded as "disclosing" amino acid sequences. The board concluded that prior art sequences do not require the filing of a sequence listing and that the Receiving Section had been wrong to apply R. 30 EPC. With reference to J. 7/11 the board pointed out that the Receiving Section is restricted to a merely formal examination of the sequence listing requirements.

The board in T. 2437/13 (coronavirus, nucleic acid, protein and methods for the generation of vaccine, medicaments and diagnostics) decided in relation to priority rights that, since the nucleic acid sequences depicted in Table 3 and sequences having at least 95% sequence identity could be generated by standard techniques, there was no need for a virus to be deposited under R. 31(1) EPC.

7.6. Deposit of living material

If an invention involves the use of or concerns biological material which is not available to the public **and** which cannot be described in the European patent application in such a manner as to enable the invention to be carried out by a person skilled in the art, the invention shall only be regarded as being disclosed as prescribed in Art. 83 EPC if a sample of the biological material has been deposited with a recognised depositary institution not later than the date of filing of the application (R. 31(1)(a) EPC) and if the application fulfils the other requirements set out in R. 31 EPC (T. 2068/11, see also G. 2/93, OJ 1995, 275). See also Guidelines F-III, 6 – March 2022 version on sufficiency of disclosure for biological material.

The disclosure of a microorganism need not depend on a deposit according to R. 28 EPC 1973 where the microorganism is sufficiently disclosed by other means (T. 2068/11, cited by T. 1338/12, which is very detailed on this issue; see also the more recent decision in T. 1045/16, relating to plant material, and T. 1376/11).

It is already the boards' firmly established case law that, where biological material is unreservedly and unrestrictedly available to the skilled person and the public – in other words, where it is sufficiently disclosed by other means – there is no need to deposit it to meet the requirements of Art. 83 EPC (T. 1338/12, point 12 of the Reasons).